AARYAN PRAVEEN

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EXPERIENCE

HumanitariansAI | Northeastern University Aug 2025 - Feb 2026
Research Assistant (Part-Time)

- Designed and implemented a browser-based traffic simulation engine in TypeScript, React, and the Canvas API, rendering 200+ concurrent vehicles while maintaining smooth real-time interactions.
- Optimized browser rendering by replacing DOM-based updates with Canvas rendering, enabling efficient visualization of hundreds of moving objects.
- Implemented realistic driver behavior models including lane-changing logic, collision avoidance, and regional traffic rules to simulate real-world freeway conditions.

Flair Labs Oct 2021 - Apr 2023
Associate Software Engineer

- Developed 20+ reusable Web Components using Lit Elements for a shared UI library adopted across three production SaaS applications, collaborating with UI/UX designers to translate Figma designs into configurable, production-ready components.
- Documented component APIs and usage guidelines to help product teams integrate reusable components consistently across multiple SaaS applications.
- Enhanced an existing Java Spring Boot and Angular application by implementing backend features, resolving production issues, and supporting deployments on AWS Elastic Beanstalk.
- Remediated 10+ security findings across six Java applications by analyzing client-provided Rapid7 and BSecure reports, implementing input validation, parameterized queries, and NGINX security policies (HSTS, CSP), then validating fixes through manual testing before client rescans.

PROJECTS

PoliSightsAI - AI-Powered Policy Intelligence Platform (Hackathon) Feb 2025 - Apr 2025

- Built an automated ETL pipeline using Apache Airflow and Selenium to scrape 100+ daily federal policies, process them with OpenAI for summarization and embeddings, then load structured data into Snowflake for analytics.
- Implemented semantic search using OpenAI embeddings and Pinecone vector database to match policies to business profiles through retrieval-augmented generation (RAG), enabling personalized recommendations based on industry and compliance requirements.
- Created a conversational interface using FastAPI and OpenAI with MongoDB backed session memory, enabling contextual follow-up querying across indexed policy datasets.

NimbleDev - Browser IDE Jan 2025 - Feb 2025

- Built a browser-based cloud development environment using React, Node.js, Docker, and WebSockets, enabling users to create and interact with isolated development workspaces in real time.
- Integrated xterm.js with node-pty to provide an interactive terminal that streamed shell input/output between the browser and backend over WebSockets.
- Automated workspace provisioning by creating Docker containers on demand, providing isolated development environments for each user session.

EDUCATION

Northeastern University Boston, MA
Master of Science in Computer Software Engineering, GPA: 3.7/4.0 May 2025
Mumbai University Mumbai
Bachelor's in Electronics Engineering, GPA: 3.2/4.0 May 2021

TECHNICAL SKILLS

Languages:	Java, JavaScript, TypeScript, Python
Frameworks & Libraries:	React, Angular, NextJS, NodeJS, ExpressJS, Spring Boot, HTML, CSS
Databases & Cloud:	MySQL, PostgreSQL, MongoDB, Pinecone, AWS (EC2, S3, Lambda, Auto Scaling Groups)
DevOps & Tools:	Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions), Git, NGINX, Figma
AI/ML:	OpenAI API, RAG, Vector Embeddings, Semantic Search, NLP, LangChain